Gaussian Differential Privacy Curve

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Definition

For $\mu \geq 0$, the Gaussian differential privacy tradeoff curve is

$$\beta_\mu(\alpha) = \Phi\left(\Phi^{-1}(1-\alpha) - \mu\right).$$

Its privacy profile is

$$\delta_\mu(\epsilon) = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - e^\epsilon \Phi\left(-\frac{\epsilon}{\mu} - \frac{\mu}{2}\right),$$

with the value at $\mu = 0$ defined by continuity as zero.

Numerical assumptions

The implementation is gated by `idealized-numerics`. Basic interval arithmetic uses the software directed backend. Native special-function results remain idealized approximations, widened outward by one native floating-point unit; this is not a certification of those special functions. Their decreasing arguments are rounded in the opposite direction before native evaluation.

Composition

Independent Gaussian privacy losses compose by addition of their squared parameters. Thus, for curves with parameters μ_i,

$$\mu_{\text{comp}} = \sqrt{\sum_i \mu_i^2}.$$

The implementation encloses each arithmetic operation with directed interval arithmetic and rejects a non-finite resulting parameter.